

Unknown side lengths



1 $AB = 18.05$

2 $AD = 37.31$

3 $AD = 49.70$

4 $h = 98, x = 28$

5

a

i $y = 90.24$

ii $w = 65.67$

iii $x = 24.6$

b

i $y = 77.80$

ii $w = 26.74$

iii $x = 51.1$

6 $CD = 6.1$

7 13 cm

8 $k = 21.76$

9

a $AB = 21.50$

b $BD = 6.3$

Unknown angles

10

a $x = 26.57^\circ$

b $y = 50.19^\circ$

c $z = 13.24^\circ$

11 $A = 66.4^\circ$

12

a $x = 29.74^\circ$

b $y = 150.26^\circ$

Applications



14

a $73\,500\text{ m}$

b $h = 41\,101\text{ m}$

15 $d = 285\text{ m}$

16

a 74.85 m

b 28.31°

17

a $\angle ADB = 23^\circ$

b $a = 35.65\text{ m}$

c $h = 20.4\text{ m}$

18 $h = 4.4\text{ m}$

19

a $x = 21.85$

b $h = 23.95$